

Bowen Feng

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Final-year MInf student at the University of Edinburgh working on the reliability of visual and multimodal AI: calibrated prediction sets, verification of vision-language answers, and audits that expose systematic failure modes.

EDUCATION

University of Edinburgh	2022 - 2027 (expected)
Master of Informatics (MInf), five-year integrated master's program, School of Informatics, Edinburgh, UK. Overall average 70/100.	
Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Discrete Mathematics and Probability, Human-Computer Interaction.	
University of California, Davis	2024 - 2025
Exchange student, Davis, CA, United States (GPA 3.82/4.0)	

RESEARCH INTERESTS

Reliability and uncertainty quantification for vision and multimodal models; verification and auditing of vision-language models and agents; selective prediction and abstention; human oversight of AI-assisted analysis.

SELECTED RESEARCH PROJECTS

Visual Verification and Prediction Sets for Vision-Language Models

First-author research | Reliability of multimodal models

- A region can change a vision-language model's answer without ruling out a plausible alternative reading. Built a four-call procedure in which a frozen model compares the target description with two rivals about the same region; the weakest margin corrects the yes/no answer. 95.23% and 96.51% accuracy on a 7,440-question AMBER attribute/relation partition on two backbones; the same rules reach F1 95.97 and 96.25 on POPE adversarial.
- Merged prediction sets from dependent visual predictors can exceed a candidate budget and lose their coverage certificate. Derived a closed-form capacity optimum that changes only label dependence and preserves every inclusion probability; usable coverage rises by 2.59 to 9.28 points across three published mergers on Office-Home and by 2.27 to 5.75 on DomainNet (28,588 images, four frozen encoders).
- Text-prompted segmenters produce plausible masks for concepts that are not in the image. FCEG-A, an area-limited foreground-context gate with a size-dependent veto bound, raises recall at a 10% false-positive rate from 68.87% to 73.92% against an ActiveSAM one-prompt adaptation on 34,702 image-disjoint test pairs, and small-object recall by 12.02 points.

TTAT: Auditable Web Workflows

First-author research | Human-computer interaction

- Developed TTAT to turn task templates into auditable web workflows, combining versioned procedures, constrained execution, and evidence tracking.
- Evaluated TTAT in a 12-participant study against conventional practice under equal model access, measuring task completion and workflow reuse.

Reliable 3D Reconstruction at Mirrors and Glass

First-author research | Computer vision

- Built CounterRay, 22 indoor scenes with 38 mirror or glass planes captured exposed, covered and from the physical side; in an audit of 200 reflective regions from ScanNet++ and ARKitScenes, 24.1% of samples accepted by released validity masks follow reflected content rather than the interface.

PUBLICATIONS

TIDE-IS: Teacher-Aligned Instance Reliability and Bounded Contour Refinement for Underwater Instance Segmentation.

Bowen Feng (sole author).
International Conference on Intelligent Sensing, Communications and Marine Applications (ISCMA 2026), accepted.

Micro-Expression Recognition System Based on Dual-Stream Graph Structure and Action Unit Auxiliary Learning.
Bowen Feng (sole author).
International Conference on Intelligent Sensing, Communications and Marine Applications (ISCMA 2026), accepted.

Probabilistic Dual-Adaptive Spatio-Temporal Graph Convolutional Networks for Forecasting Energy Consumption Dynamics of Electric Vehicle Charging Stations.
Djamel Eddine Mekkaoui, Mohamed Amine Midoun, Abdelkarim Smaili, **Bowen Feng**, Mohamed Zakariya Talhaoui, and Yanming Shen.
Computers & Electrical Engineering, 122, 109976 (2025). doi.org/10.1016/j.compeleceng.2024.109976

SPR: A Similar Projection Revisor for Complex Logical Reasoning over Knowledge Graphs.
Tingting Wang, Yuxuan Tang, Ruolin Li, Duo Yu, **Bowen Feng**, Feng Ding, Shuo Yu, and Yanming Shen.
Web Information Systems and Applications (WISA 2024), LNCS 14883, pp. 78-90. doi.org/10.1007/978-981-97-7707-5_7

UNDER REVIEW

When Should a Text-Prompted Segmenter Abstain? Area-Limited Foreground-Context Evidence for One-Prompt Presence Decisions.
Bowen Feng. Under review (journal), 2026.

One-Sided Residual Certificates for Exact Quantized Image Retrieval.
Bowen Feng. Under review (journal), 2026.

MANUSCRIPTS IN PREPARATION

Rivals Before Votes: Four-Call Visual Verification with Shared-Witness Competition First author. Manuscript in preparation, 2026.

Closed-Form Capacity Gains for Reliable Visual Set Merging First author. Manuscript in preparation, 2026.

RESEARCH EXPERIENCE

Research with Prof. Tian Xie, Zhejiang University	2025 - 2026
Auditable web workflows (TTAT): system design, WebArena evaluation and a 12-user study.	
Research assistant, Prof. Yanming Shen's group, Dalian University of Technology	2022 - 2024
Knowledge-graph reasoning (WISA 2024) and spatio-temporal forecasting (Computers & Electrical Engineering 2025): data processing, temporal modeling components and baseline comparisons.	
Bioinformatics and Statistical Analysis Intern, CAACT, Beijing	2025
Prepared data and covariates in R and Python; model selection, multiple-comparison control, effect sizes and confidence intervals; sensitivity analyses and reproducible reports.	

SERVICE

Student Representative (Year 4), School of Informatics, University of Edinburgh	2025 - 2026
Programme Representative, Computer Science, School of Informatics, University of Edinburgh	2023 - 2024
Year 2 Student Representative, School of Informatics, University of Edinburgh	2023 - 2024

SKILLS

Programming: Python, R, C++, JavaScript, LaTeX.

Frameworks and tools: PyTorch, OpenCV, NumPy/SciPy, scikit-learn, pandas, Playwright and Selenium for browser automation, Git; training on multi-GPU clusters.

Methods: Controlled and counterfactual evaluation, paired-capture datasets, calibration and conformal prediction, ablation studies, user studies with statistical inference, failure analysis.